



C.2 Wave model

Wave behaviour · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A travelling wave transfers energy through a medium (or through space, for electromagnetic waves) without net transfer of matter, characterised by wavelength λ , frequency f , period $T = 1/f$, and wave speed $v = f\lambda$. Transverse waves have particle oscillation perpendicular to the direction of energy transfer (e.g. light, waves on a string), while longitudinal waves have oscillation parallel to it (e.g. sound). Wavefronts are surfaces of constant phase, and intensity (power per unit area) for a point source spreading spherically follows the inverse square law, $I \propto 1/r^2$. The electromagnetic spectrum spans radio, microwave, infrared, visible, ultraviolet, X-ray and gamma-ray radiation, all travelling at speed c in a vacuum but differing in wavelength/frequency.

Worked example: follow the reasoning

A wave source of power 8.0 W emits sound uniformly in all directions. Find the intensity at a distance of 4.0 m from the source.

$$I = P/(4\pi r^2) = 8.0/(4\pi(4.0)^2) = 8.0/(4\pi \times 16) = 8.0/201.1 = 0.0398 \text{ W}\cdot\text{m}^{-2}.$$

Answer: $I \approx 0.040 \text{ W}\cdot\text{m}^{-2}$.

Quick check · multiple choice

A wave has frequency 250 Hz and travels at $340 \text{ m}\cdot\text{s}^{-1}$ through air. What is its wavelength?

- A 0.73 m
- B 1.36 m
- C $8.5 \times 10^4 \text{ m}$
- D 85000 m

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A point source radiates sound uniformly in all directions with total power 20 W. [7]
- (a) Calculate the intensity of the sound at a distance of 5.0 m from the source.
 - (b) Calculate the distance from the source at which the intensity falls to $0.010 \text{ W}\cdot\text{m}^{-2}$.
 - (c) State how the amplitude of the sound wave varies with distance from the source, given that intensity is proportional to amplitude squared.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $I = P/(4\pi r^2) = 20/(4\pi(5.0)^2) = 20/(4\pi \times 25) = 20/314.2 = 0.0637 \text{ W}\cdot\text{m}^{-2}$.

(b) $0.010 = 20/(4\pi r^2) \rightarrow r^2 = 20/(4\pi \times 0.010) = 20/0.1257 = 159.2 \rightarrow r = 12.6 \text{ m}$.

(c) Since $I \propto 1/r^2$ and $I \propto \text{amplitude}^2$, $\text{amplitude} \propto 1/r$ — amplitude decreases in proportion to $1/r$, i.e. inversely with distance.

Quick check answer

B. $\lambda = v/f = 340/250 = 1.36 \text{ m}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.